

Groundwater quality assessment using Water Quality Index (WQI) in parts of Anantapur, A.P. (India)

P. Ravi Kumar, S. Srinivasa Gowd*, C. Krupavathi and B. Pradeep Kumar

Department of Geology, Yogi Vemana University, YSR Kadapa, AP, 516005, India.

*Corresponding author: ssgowd@gmail.com

ABSTRACT

Groundwater is one of the important freshwater resources available for human utilization. Assessment of the groundwater quality is essential for improving its portability. The utilization of groundwater has been greater than ever before during recent times. The present work is intended at evaluating the water quality index (WQI) in Ramagiri Mandal of Anantapur district (Andhra Pradesh). Twenty-two groundwater samples were collected from various sites and WQI for various parameters such as total hardness, chloride, electrical conductivity, total dissolved solids, major anions CO_3^{2-} , HCO_3^- , Cl^- and major cations like Ca^{2+} , Mg^{2+} , Na^+ , K^+ are evaluated. These parameters were considered to calculate the water quality index, which ranges between 75.61 to 109.33. In most part of the studied region, water is appropriate for drinking purposes. WQI data reveals that 86% of groundwater samples fall into good water quality while the remaining 14% fall into the very poor water category. This study will contribute to our understanding of future WQI planning and research for drinking water, irrigation, residential, and industrial applications.

Keywords: Groundwater, Hydrogeochemistry, Ramagiri, WHO, Water quality index (WQI)

INTRODUCTION

Freshwater is scarce, yet demand is growing day by day. Groundwater consumption is critical when surface water is not available in sufficient quantity. Groundwater is a sustainable natural resource that is replenished by precipitation on an annual basis. The protection and conservation of groundwater quality are dependent on its quality. As a result, it's critical to examine groundwater quality not just for its current usage, but also as a possible supply of water for future use (Gowd 2005; Adimalla et al., 2018; Zhang et al., 2020; Bahir et al., 2020; Golla et al., 2021). Groundwater plays a dynamic role in the world economy. About 64% of rural and 37% municipal population in India are without the availability of safe drinking water as per the WHO. Life is not possible without water and thus considered as the most indispensable resource (Ramakrishnaiah et al., 2009; Seifi et al., 2020). It is also utilized in industry and agriculture. This study used GIS as a database structure in order to prepare maps of water quality, which is vital to estimate the water safeness for drinking and agriculture reasons and also as a deterrent warning to the environmental potential and physical condition (Gowd et al., 2005; Krupavathi et al., 2020).

Chemical properties of groundwater are significant in identifying and assessing water quality. It helps to identify potential changes in quality. Groundwater quality is determined through a variety of chemical components and their concentrations, which are largely obtained from geological data in the area (Aryal et al., 2012; Balaji et al., 2017; Karunanidhi et al., 2020; Ram et al., 2021; Reddy et al., 2022). The present study uses geospatial techniques in groundwater quality index mapping. In this study area, the

main objective of research work is to assess the water quality index using Geospatial techniques. Geospatial techniques have emerged as efficient powerful tool in different fields of science and industrial purpose (Babiker et al., 2007; Kumar et al., 2019; Badapalli et al., 2021). It enables us to acquire data that is referenced to the earth and used for analysis, modeling, simulations, and visualization. The water quality index (WQI) is one of the most important tools to communicate information on the quality of water to concerned citizens and policymakers (Vasanthavigar et al., 2010).

STUDY AREA

The study area of Ramagiri Mandal is situated in the southern corner of Anantapur district in the state of Andhra Pradesh (India) and occupy an area of 339 sq. km. Anantapur district, located in the rain shadow of the Western Ghats and the interior of the Deccan Plateau, is one of the country's most frequently drought-affected district, with an average annual rainfall of about 520 mm, the district is the driest in the state and, after Jaisalmer, the second driest in the country (Rajasekhar et al., 2019; Kumar et al., 2020). Anantapur has a tropical climate, with temperatures ranging from 24 to 46° C throughout the summer months of March to May. The majority of people in Anantapur rely on agriculture for their living. The study area located in between 14° 18' 0" and 14° 24' 0" N and 77° 30' 0" and 77° 36' 0" E with a mean elevation of 516 meters (Figure 1). Ramagiri Mandal falls in the survey of India (SOI) topographic sheet 57 F/7 and 57F/11. The study area lithologically consists of hornblende biotite gneiss, meta basalts, grey/pink granites, granites, granodiorite, amphibolite, and quartzite (Figure 2).

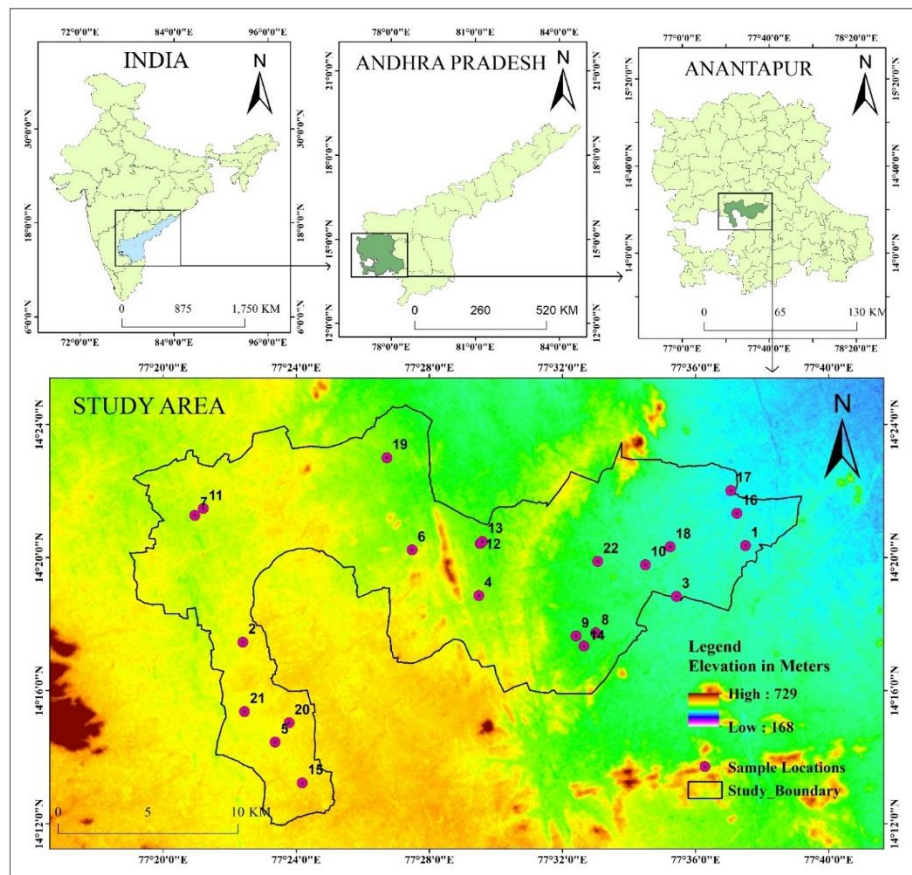


Figure 1. Sample location map in the study area, located in the Ramagiri mandal of Anantapur district (Andhra Pradesh).

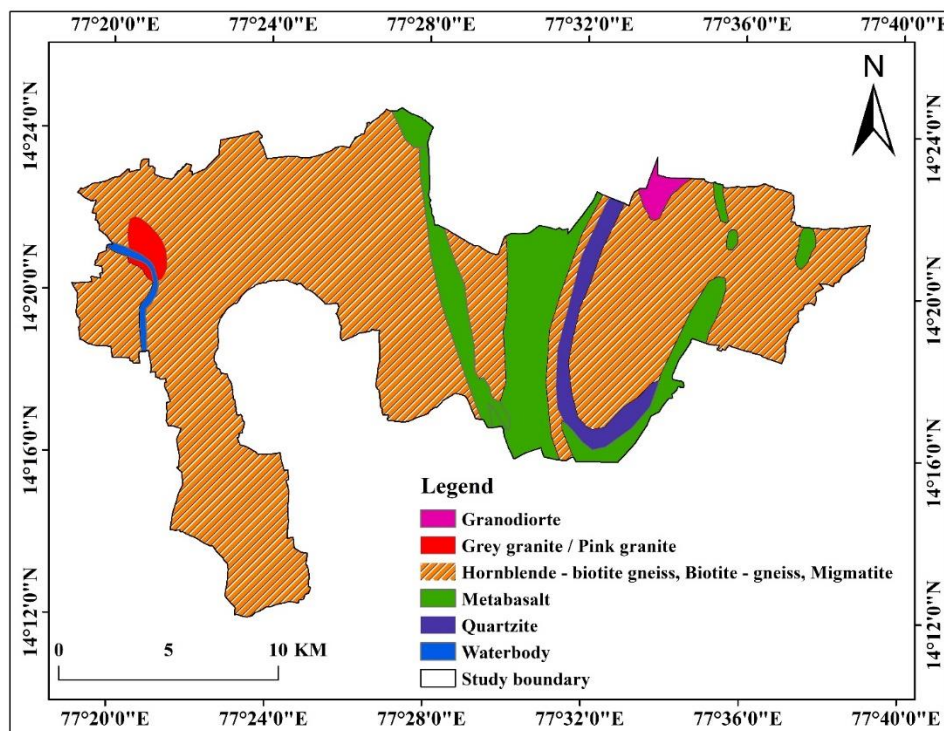


Figure 2. Geological map of the study area.

METHODOLOGY

Twenty-two groundwater samples were collected from open and dug wells at different locations of the Ramagiri mandal for the period of post-monsoon in January 2021. The groundwater samples, collected in well-cleaned one ltr polyethylene bottles, have been analysed in the laboratory and the results are shown in Tables 1 and 2, which show physiochemical parameter's statistical summary. Table 3

shows comparison between groundwater quality drinking water standards, WHO. The latitude and longitude of the sample locations are identified with a Global positioning system. The water quality index was measured by choosing appropriate water quality parameters, which play a vital role in determining water quality as per the weighted arithmetic mean method after giving appropriate weightage to the selected parameters (Ramakrishnaiah et al., 2009). The adopted methodology is shown in the Figure 3.

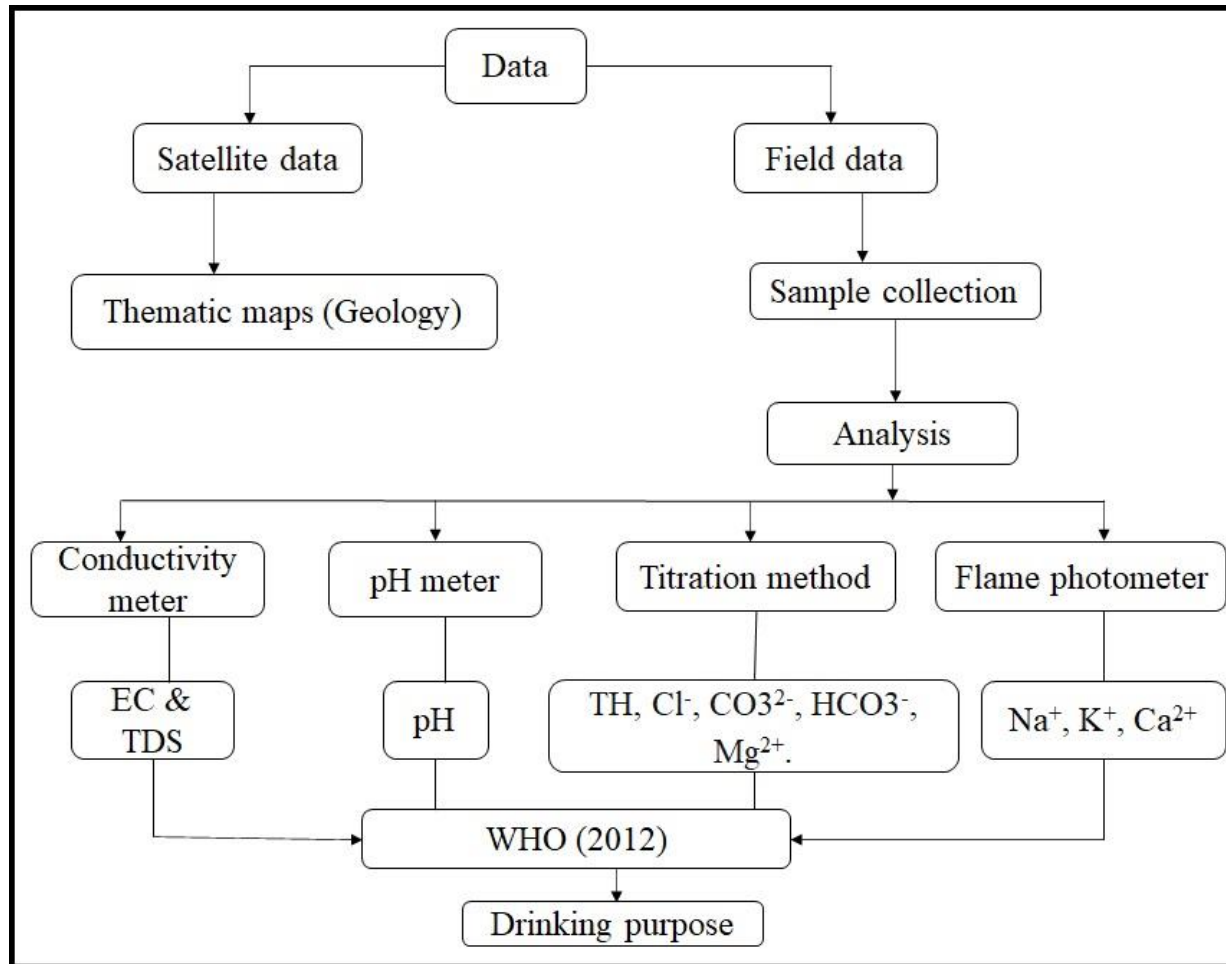


Figure 3. Methodology adopted in the analysis of the samples from the study area located in Ramagiri mandal, Anantapur district (Andhra Pradesh).

Table 1. Analytical data for groundwater samples and their locations.

Sample no	Sample locations	Latitude	Longitude	TH	Cl ⁻	CO ₃ ²⁻	HCO ₃ ⁻	pH	EC	TDS	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺
				Mg/l	Mg/l	Mg/l	Mg/l		μS/cm	Mg/l	Mg/l	Mg/l	Mg/l	Mg/l
1	Ganthimarri	N 14° 20' 14"	E 77° 37' 36"	120	385	26	110	8	1034	970	85	24	162	8
2	Kondapuram	N 14° 17' 25"	E 77° 22' 23"	180	275	45	124	7	1643	1860	160	42	125	6
3	Kuntimaddi	N 14° 18' 40"	E 77° 35' 33"	210	228	34	61	7	1835	790	108	36	135	5
4	Makkinavaripalle	N 14° 18' 33"	E 77° 22' 25"	150	280	24	98	8	1927	954	94	42	125	7
5	Motarchinthalapalle	N 14° 14' 30"	E 77° 23' 47"	240	340	52	146	7	1532	1015	92	65	148	4
6	Nasanakota	N 14° 20' 04"	E 77° 25' 58"	160	220	22	116	7	3122	1255	70	23	125	5
7	Perur	N 14° 21' 04"	E 77° 20' 58"	130	310	58	108	7	1892	987	150	65	178	6
8	Polepalli	N 14° 18' 42"	E 77° 33' 55"	120	260	36	135	7	3093	1740	96	24	138	5
9	Ramagiri	N14°18'37.4"	E77°30'23"	320	218	28	156	8	3900	1964	114	26	125	5
10	Seshadribatrahalli	N 14° 19' 41"	E 77° 34' 30"	160	276	45	160	8	2850	1425	130	18	168	6
11	Reddivaripalle	N 14° 21' 24"	E 77° 29' 35"	210	295	30	194	8	2612	1516	128	56	228	7
12	R. Kothapalli	N14°20'24"	E 77° 21' 06"	140	367	20	71	8	3100	1498	90	24	148	6
13	Venkatapuram	N 14° 21' 42"	E 77° 23' 36"	134	248	18	84	7	991	462	90	19	117	5
14	Akkampalli	N 14° 17' 55"	E 77° 30' 19"	120	289	36	156	7	976	492	108	49	142	6
15	Garimekalapalli	N 14° 13' 21"	E 77° 24' 08"	190	384	20	118	7	1960	898	88	28	128	7
16	Penubole	N 14° 21' 29"	E 77° 37' 09"	230	219	36	91	8	1873	947	78	34	118	5
17	Kalikivandlapalli	N 14° 21' 58"	E 77° 36' 54"	340	396	24	121	8	1560	862	106	44	158	2
18	Madapuram	N 14° 23' 00"	E 77° 26' 44"	260	208	20	152	8	987	435	94	28	268	6
19	Suddakuntapalli	N 14° 20' 19"	E 77° 35' 15"	150	345	64	165	8	1264	685	124	45	288	7
20	Mangapuram	N 14° 27' 25"	E 77° 26' 35"	180	256	54	142	8	1814	978	92	24	198	5
21	Thimmapuram	N 14° 25' 43"	E 77° 26' 30"	140	285	46	186	7	1085	612	86	34	284	6
22	Edugurallapalli	N 14° 19' 53"	E 77° 33' 55"	100	410	58	198	7	2185	1288	124	61	286	5

Table 2. Statistical analysis of physiochemical parameters of the study area.

Parameters	Minimum	Maximum	average	Stdv	Cv %
TH	100	340	181.1	64.43	35.577
Cl ⁻	208	410	295.1	63.56	21.5385
CO ₃ ²⁻	18	64	36.1	14.43	39.9723
HCO ₃ ⁻	61	198	131.4	38.07	28.9726
pH	7.1	8.1	7.5	1.3	17.3333
EC	976	3900	1965.2	828.64	42.1657
TDS	435	1964	1074	443.73	41.3156
Ca ²⁺	70	160	105	23.03	21.9333
Mg ²⁺	18	65	37	14.91	40.2973
Na ⁺	117	288	172	59.22	34.4302
K ⁺	2	8	5.6	1.25	22.3214

Table 3. Comparison of groundwater quality with standard of drinking water (WHO, 2012).

Water quality parameter	WHO maximum acceptable limit.	WHO minimum acceptable limit.	Concentration in the study area
TH (mg/l)	500	100	100-340
Cl ⁻ (mg/l)	600	200	208-410
pH	8.5	7	7.1-8.1
EC (μS/cm)	1500	400	976-3900
Ca ²⁺ (mg/l)	200	75	70-160
Mg ²⁺ (mg/l)	150	50	18-65

Table 4. Correlation analysis between physicochemical parameters of the study area

Chemical parameter	TH	Cl ⁻	CO ₃ ²⁻	HCO ₃ ⁻	pH	EC	TDS	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺
TH	1.00										
Cl ⁻	-0.23	1.00									
CO ₃ ²⁻	-0.36	0.26	1.00								
HCO ₃ ⁻	-0.10	0.22	0.53	1.00							
pH	0.29	-0.24	-0.27	-0.05	1.00						
EC	0.19	-0.20	-0.22	-0.05	0.22	1.00					
TDS	0.11	-0.03	-0.01	0.09	0.16	0.81	1.00				
Ca ²⁺	0.00	0.08	0.35	0.22	-0.17	0.08	0.36	1.00			
Mg ²⁺	-0.07	0.42	0.56	0.39	-0.48	-0.26	-0.07	0.42	1.00		
Na ⁺	-0.18	0.27	0.44	0.70	0.05	-0.29	-0.25	0.15	0.32	1.00	
K ⁺	-0.42	-0.01	-0.14	-0.01	0.20	-0.07	-0.02	0.14	-0.16	0.13	1.00

Table 5. Relative weights for each parameter.

Chemical parameters	Sn	1/Sn	K	Wn= K/Sn	Ideal value (Vi)	Observed value (Vn)	Vn/Sn	Vn/Sn* 100=Qn	WnQn	WQI
pH	8.5	0.11	5.2	0.6154	7	7.6	0.4	40	24.61	54.7
TH	300	0.003	5.2	0.0174	0	120	0.4	40	0.69	
Cl ⁻	250	0.004	5.2	0.0209	0	385	1.54	154	3.22	
EC	2000	0.0005	5.2	0.0026	0	1034	0.517	51.7	0.13	
TDS	1500	0.0007	5.2	0.0034	0	970	0.646	64.66	0.22	
Ca ²⁺	75	0.013	5.2	0.0697	0	85	1.13	113.33	7.9	
Mg ²⁺	30	0.03	5.2	0.1743	0	24	0.8	80	13.95	
Na ⁺	150	0.007	5.2	0.0348	0	162	1.08	108	3.76	
K ⁺	150	0.007	5.2	0.0348	0	8	0.053	5.33	0.18	

Correlation matrix

A high positive correlation was found between TDS and EC (0.814), bicarbonates and sodium (0.697), calcium and magnesium (0.42), calcium and TDS (0.363). While the

negatively correlated values were found between TH and potassium (-0.422) and magnesium and pH (-0.4796) and carbonate and pH (-0.27) as shown in Table 4.

Water quality index (WQI) analysis

The water quality Index (WQI) uses the World Health Organization (WHO) and ISI recommended drinking water quality standards. The WQI calculation was carried out using a weighted arithmetic index as shown below. Water

quality index of relative weights for each parameter are given in Table 5 and WQI at individual sampling location were shown in Table 6. According to the water quality index classification, 86 % of the groundwater samples fall into good and the remaining 14 %, fall in poor water category, as shown in Table 7.

Table 6. WQI at individual sampling locations

Sample no	Sample locations	Latitude	Longitude	WQI	Water Quality Status
1	Ganthimarri	N 14° 20' 14"	E 77° 37' 36"	99.91	Good Water
2	Kondapuram	N14° 17' 25"	E 77° 22' 23"	89.91	Good Water
3	Kuntimaddi	N 14° 18' 40"	E 77° 35' 33"	89.91	Good Water
4	Makkinavaripalle	N 14° 18' 33"	E 77° 22' 25"	97.11	Good Water
5	Motarchinthalapalle	N 14° 14' 30"	E 77° 23' 47"	103.56	Poor Water
6	Nasanakota	N 14° 20' 04"	E 77° 25' 58"	79.23	Good Water
7	Perur	N 14° 21' 04"	E 77° 20' 58"	85.58	Good Water
8	Polepalli	N 14° 18' 42"	E 77° 33' 55"	83.35	Good Water
9	Ramagiri	N14° 18' 37.4"	E77° 30' 23"	81.73	Good Water
10	Seshadribatrahalli	N 14° 19' 41"	E 77° 34' 30"	81.95	Good Water
11	Reddivaripalle	N14° 21' 24"	E 77° 29' 35"	109.33	Poor Water
12	R. Kothapalli	N14° 20' 24"	E 77° 21' 06"	86.29	Good Water
13	Venkatapuram	N 14° 21' 42"	E 77° 23' 36"	78.14	Good Water
14	Akkampalli	N 14° 17' 55"	E 77° 30' 19"	97.41	Good Water
15	Garimekalapalli	N 14° 13' 21"	E 77° 24' 08"	85.24	Good Water
16	Penubole	N 14° 21' 29"	E 77° 37' 09"	88.54	Good Water
17	Kalikivandlapalli	N 14° 21' 58"	E 77° 36' 54"	99.2	Good Water
18	Madapuram	N14° 23' 00"	E 77° 26' 44"	90.62	Good Water
19	Suddakuntapalli	N 14° 20' 19"	E 77° 35' 15"	103.66	Poor Water
20	Mangapuram	N 14° 27' 25"	E 77° 26' 35"	89.53	Good Water
21	Thimmapuram	N 14° 25' 43"	E 77° 26' 30"	91.57	Good Water
22	Edugurallapalli	N14° 19' 53"	E 77° 33' 55"	75.61	Good Water

Table 7. Classifications of waterquality types based on WQI (Ramakrishnaiah et al., 2009)

Range	Water type	Number of villages in the study area
<50	Excellent	Nil
50-100	Good	19
100-200	Poor	3
200-300	Very poor	Nil
>300	Water unsuitable for drinking	Nil

Water quality (QN) calculation

Water quality (qn) has been calculated using following equation

$$q_n = 100 [(V_n - V_i) / (S_n - V_i)] \dots\dots\dots (1)$$

where

q_n = nth parameter of water quality rating,

V_n = the value of nth parameter observed,

S_n = Standard acceptable value of the nth parameter,

V_i = Ideal value for the nth parameter.

All ideal values (V_i) are taken as zero for drinking water except pH 7.0.

Unit weight Calculation (W_n):

The Unit weight of the corresponding parameter was calculated by an inverse proportional value to the recommended standard value S_n .

$$W_n = K / S_n \dots\dots\dots (2)$$

where

W_n = n^{th} Parameter of unit weight,

S_n = n^{th} parameter standard value,

K = constant proportionality,

The proportionality constant (K) has been calculated here using the equation:

$$K = 1 / \sum (1/S_n) \dots\dots\dots (3)$$

The total water quality index (WQI) was then calculated linearly by adding the quality assessment to the unit weight.

$$WQI = \sum W_n Q_n / \sum W_n \dots\dots\dots (4)$$

GROUNDWATER CHEMISTRY

pH is a term used universally to express the intensity of the acid or alkaline condition of a solution. It is a very important parameter in groundwater media to provide an important piece of information of geochemical equilibrium.

The maximum pH was noted as 8.1 at sampling location Mangapuram and the minimum was 7.5 at Peruru village. When compared with standard values of WHO (2012) 6.5-8.5, it was found to be within the permissible limit at all locations. The pH of the study area belongs to alkaline nature. pH values are important in medicine, living sciences, and many other applications (Solangi et al., 2017, WHO, 2021).

Total hardness

Total hardness is described as the number of suspended materials of magnesium and calcium in the water (Balakrishnan et al., 2011). Total hardness in the study region ranges from 100 to 340 mg/l (Table1) (ISI et al., 2012). Todd classification on Total Hardness of ground water has been shown in the Table 8. Water hardness is chiefly due to the result of contact between water and geological formations. In the study area, the minimum value of TH was noted at Edugurallapalli and maximum concentration was observed at Kalikivandlapalli.

Total dissolved solids (TDS)

A total dissolved solid is utilized to explain the inorganic salts and little quantity of organic matter present in the sample. In the study region, the value of TDS ranges from 462 to 1964 mg/L (Table 1). Total dissolved solids in drinking water should be less than 500 ppm (Aghazadeh et al., 2017). In the study area, minimum TDS range occurred in sample no 13 and maximum concentrations in sample number 9. At various places, TDS concentrations exceeded the highest useful limit and are therefore unsuitable for drinking purposes.

Chloride

Chloride naturally occurs in water and is found as a component of salt or in some cases in a mixture of potassium and calcium (Chidambaram et al., 2014). Chloride concentration range from 208 mg/l to 396 mg/l. Minimum concentrations were observed at Mangapuram and high concentration, at Edugurallapalli during the post-monsoon period. The WHO standard permissible limit of chloride concentration for drinking water is specified as 250 mg/l to 600mg/l, compared with WHO standard, chloride concentrations post-monsoon observed all groundwater samples occurred below the required limit.

Table 8. Classification of groundwater based on Total Hardness

Parameter	range	Classification
Total Hardness	Safe	<75
	Moderate hard	75-150
	Hard	150-300
	Very hard	>300

Electrical conductivity

EC is hazard to crops. As the concentration of dissolved salts increases, it gradually increases conductivity too (Ibraheem and Khan, 2017; Nagaraju et al., 2014). In the study area, observed EC ranges from 976 $\mu\text{S}/\text{cm}$ to 3900 $\mu\text{S}/\text{cm}$. The minimum value is observed in Cherlopalli and the highest value in Ramagiri village (Table 1).

Total Alkalinity (CO_3^- and HCO_3^-)

Alkalinity is a water acid-neutralizing capability and is the principal function of carbonate and hydroxide content (Islam et al., 2014). The bicarbonate concentration of the study area varied from 61 to 198 mg/l. Minimum concentration was noticed in Kuntimaddi and high concentration was noticed at Thimmapuram (Table 1). The WHO (2012) standard allowable limit of bicarbonate concentration for drinking water is specified as 300 mg/l to 600mg/l. When compared with WHO standard, HCO_3^- concentrations in all samples are seen to fall within the acceptable limit.

Calcium and Magnesium

Calcium occurs naturally in water. Its main source is geological formations like limestone, calcite, marble, gypsum, and fluorite (Sivasubramanian et al., 2013). The concentration of the calcium ion was found to vary widely from 78 to 160 mg/l (Table 1). Lower concentrations of 78 mg/l are measured at Akkampalli and the maximum 160 mg/l at Kondapuram. As per WHO (2012), acceptable limit for drinking water is 75 to 200 mg/l, values at all locations are within the acceptable limit. Similarly, magnesium varied from 18 to 65 mg/l. Low concentration was observed at Seshadribatrahalli and higher value was noted at Motarchinthalapalle. The WHO (2012) limit for drinking water is 30 to 150 mg/l, and thus all the studied samples were observed to be within the allowable limit.

Sodium

Sodium is an abundant element on Earth and is broadly dispersed in soils, plants, water. Sodium-ion is everywhere

in water because of the high solubility of sodium salts (Kumar et al., 2020). The Sodium concentration varied from 117 mg/l at Venkatapuram to a maximum value 288 mg/l at P. Kottala. The WHO (2012) standard for drinking water is 50mg/l to 200mg/l. In the study area 17 samples were within the permissible limit while remaining 7 samples are below the desired limit.

Potassium

The concentration of potassium is mostly less in groundwater. In the study area, it varied from 2mg/l at Edugurallapalli to 8.0 mg/l at Ganthimarri (Table 1). The WHO (2012) standard for drinking water permissible limit is 10 to 12 mg/l. All the samples are within the allowable limit.

CONCLUSIONS

The present study aimed to assess the water quality index of groundwater samples from Ramagiri mandal, Anantapur district (Andhra Pradesh). In the study area, at least 70% of the population is dependent on groundwater sources for drinking purposes. According to the investigations, some water samples of Ramagiri mandal were observed within the permissible limit, while some others have above the permissible limit as the area is growing with lot of agricultural activities and continuous disposal of municipal wastes without proper treatment. It is recommended to control and restrict the excess usage of fertilizer for agricultural activities in the mandal to prevent the leachates from reaching the groundwater. Twenty-two groundwater samples were collected from various sites and WQI was estimated including various parameters like total hardness, chloride, electrical conductivity, total dissolved solids, major anions CO_3^{2-} , HCO_3^- , Cl^- and major cations Ca^{2+} , Mg^{2+} , Na^+ , K^+ . These parameters were considered for calculating the water quality index, which ranges between 75.61 and 109.33. Most of this study region is appropriate for drinking purposes. WQI data reveals that 86% of groundwater samples fall into good water, while the remaining 14% fall into the very poor category.

ACKNOWLEDGEMENTS

The first author P. Ravi Kumar is greatly thankful to the Department of Science and Technology (DST), Government of India, for financial support through Inspire program (Sanction order No. DST/INSPIRE Fellowship/2018/IF180877) and also thankful to the Department of Geology, Yogi Vemana University, for necessary facilities for carrying out my research work.

Compliance with Ethical Standards

The authors declare that they have no conflict of interest and adhere to copy right norms.

REFERENCES

- Adimalla, N., Li, P. and Venkatayogi, S., 2018. Hydrogeochemical evaluation of groundwater quality for drinking and irrigation purposes and integrated interpretation with water quality index studies. *Environ. Proc.*, 5(2), 363-383.
- Aghazadeh, N., Chitsazan, M. and Golestan, Y., 2017. Hydrochemistry and quality assessment of groundwater in the Ardabil area, Iran. *Applied Water Sci.*, 7(7), 3599-3616.
- Aryal, J., Gautam, B. and Sapkota, N., 2012. Drinking water quality assessment. *J. Nepal Health Res. Council*, 10(22), 192-196.
- Babiker, I.S., Mohamed, M.A. and Hiyama, T., 2007. Assessing groundwater quality using GIS. *Water Res. Management*, 21(4), 699-715.
- Badapalli, P.K., Kottala, R.B., Sree, P.P. and Rajasekhar, M., 2021. Occurrence and structures of dolomites in North Eastern part of Anantapur district, and their use in engineering materials. *Materials Today Proc.*, 50, 1005-1010.
- Bahir, M., Ouhamdouch, S., Ouazar, D. and Chehbouni, A., 2020. Assessment of groundwater quality from semi-arid area for drinking purpose using statistical, water quality index (WQI) and GIS technique. *Carbonates and Evaporites*, 35(1), 1-24.
- Balaji, E., Nagaraju, A., Sreedhar, Y., Thejaswi, A. and Sharifi, Z., 2017. Hydrochemical characterization of groundwater in around Tirupati Area, Chittoor District, Andhra Pradesh, South India. *Applied Water Sci.*, 7(3), 1203-1212.
- Balakrishnan, P., Saleem, A. and Mallikarjun, N.D., 2011. Groundwater quality mapping using geographic information system (GIS): A case study of Gulbarga City, Karnataka, India. *African J. Environ. Sci. and Tech.*, 5(12), 1069-1084.
- Chidambaram, S., Anandhan, P., Prasanna, M.V., Thivya, C., Thilagavathi, R. and Sarathidasan, J., 2014. Determination of the utility of groundwater with respect to the geochemical parameters: a case study from Tuticorin District of Tamil Nadu (India). *Environ. Develop. sustain.*, 16(3), 689-721.
- Ibraheem, A.M. and Khan, S.M.M.N., 2017. Suitability assessment of groundwater for irrigation purpose in Veppanthattai block, Perambalur district, Tamil Nadu. *World Scientific News*, 81(2), 81-93.
- ISI, 2012. Drinking water standards, Table 1, Substance and characteristics affecting the acceptability of water for domestic use. 18, 10500. Ind. Std. Instt., New Delhi.
- Islam, M.J., Hakim, M.A., Hanafi, M.M., Juraimi, A.S., Aktar, S., Siddiqua, A., Rahman, A.S., Islam, M.A. and Halim, M.A., 2014. Hydrogeochemical quality and suitability studies of groundwater in northern Bangladesh. *J. Environ. Bio.*, 35(4), 765.
- Golla, V., Badapalli, P.K. and Mannala, P., 2021. Assessment of groundwater quality for drinking and irrigation in semi-arid regions of Andhra Pradesh, Southern India, using multivariate statistical analysis. *Arabian J. Geosci.*, 14(19), 1-11.
- Gowd, S.S., 2005. Assessment of groundwater quality for drinking and irrigation purposes: a case study of Peddavanka watershed, Anantapur District, Andhra Pradesh, India. *Environ. Geology*, 48(6), 702-712.
- Gowd, S.S., Krishna, A.K. and Govil, P.K., 2005. Environmental risk assessment and remediation of soils contaminated due to waste disposal from tannery industries: A case study of Ranipet industrial area, Tamil Nadu, India. *Geochim. Cosmochim. Acta Suppl.*, 69(10), p. A427.
- Karunanidhi, D., Aravinthasamy, P., Deepali, M., Subramani, T. and Roy, P.D., 2020. The effects of geochemical processes on groundwater chemistry and the health risks associated with fluoride intake in a semi-arid region of South India. *RSC Advances*, 10(8), 4840-4859.
- Krupavathi, C., Gowd, S.S., Maheswararao, R., Kumar, P.R., and Kumar, B.P., 2020. Analysis of Groundwater quality using water quality index: A case study of Simhadripuram Mandal, YSR Kadapa District, Andhra Pradesh. *The Intl. J. Analyt. Experim. modal analy.* No.0886-9367.
- Kumar, B.P., Babu, K.R., Rajasekhar, M. and Ramachandra, M., 2019. Assessment of land degradation and desertification due to migration of sand and sand dunes in Beluguppa Mandal of Anantapur district (AP, India), using remote sensing and GIS techniques. *J. Ind. Geophys. Union*, 23(2), 173-180.
- Kumar, P.R., Gowd, S.S., Kumar, B.P., Maheswararao, R. and Krupavathi, C., 2020. Hydro geochemical Studies of Nambulapulakunta Mandal, Anantapur District, Andhra Pradesh by using Geospatial techniques. *The Intl. J. Analyt. Experim. modal analy.* No.0886-9367.
- Nagaraju, A., Sreedhar, Y., Kumar, K.S., Thejaswi, A. and Sharifi, Z., 2014. Assessment of groundwater quality and evolution of hydrochemical facies around Tummalapalle area, Cuddapah District, Andhra Pradesh, South India. *J. Environ. Analyt. Chem.*, 1(2), 1-6.
- Rajasekhar, M., Raju, G.S., Sreenivasulu, Y., and Raju, R.S., 2019. Delineation of groundwater potential zones in semi-arid region of Jilledubanderu river basin, Anantapur District, Andhra Pradesh, India using fuzzy logic, AHP and integrated fuzzy-AHP approaches. *HydroRes.*, 2, 97-108.
- Ram, A., Tiwari, S.K., Pandey, H.K., Chaurasia, A.K., Singh, S. and Singh, Y.V., 2021. Groundwater quality assessment using water quality index (WQI) under GIS framework. *Appl. Water Sci.*, 11(2), 1-20.
- Ramakrishnaiah, C.R., Sadashivaiah, C. and Ranganna, G., 2009. Remote sensing and GIS techniques for evaluation of groundwater quality in Tumkur taluk, Karnataka State, India. *Intl. J. Appl. Environ. Sci.*, 4(2), 137-150.
- Reddy, C.K.V., Golla, V., Badapalli, P.K. and Reddy, N.B.Y., 2022. Evaluation of groundwater contamination for fluoride and nitrate in Nellore Urban Province, Southern

- India: a special emphasis on human health risk assessment (HHRA). *Appl. Water Sci.*, 12(3), 1-9.
- Seifi, A., Dehghani, M., and Singh, V.P., 2020. Uncertainty analysis of water quality index (WQI) for groundwater quality evaluation: Application of Monte-Carlo method for weight allocation. *Ecolog. Indic.*, 117, 106653.
- Solangi, G.S., Siyal, A.A., Babar, M.M. and Siyal, P., 2017. Groundwater quality mapping using geographic information system: A case study of District Thatta, Sindh. *Mehran Univ. Res. J. Engg. Tech.*, 36(4), p.14.
- Sivasubramanian, P., Balasubramanian, N., Soundranayagam, J.P. and Chandrasekar, N., 2013. Hydrochemical characteristics of coastal aquifers of Kadaladi, Ramanathapuram District, Tamilnadu, India. *Appl. Water Sci.* 3(3), 603-612.
- Vasanthavignar, M., Srinivasamoorthy, K., Vijayaragavan, K., Rajiv Ganthi, R., Chidambaram, S., Anandhan, P., Manivannan, R. and Vasudevan, S., 2010. Application of water quality index for groundwater quality assessment: Thirumanimuttar sub-basin, Tamilnadu, India. *Environ. Monit. Assess.* 171(1), 595-609.
- WHO (World Health Organization) 2012. Good health adds life to years: Global brief for World Health Day 2012 (No. WHO/DCO/WHO/2012.2). World Health Organization.
- WHO (World Health Organization) 2021. A global overview of national regulations and standards for drinking-water quality.
- Zhang, Q., Xu, P. and Qian, H., 2020. Groundwater quality assessment using improved water quality index (WQI) and human health risk (HHR) evaluation in a semi-arid region of northwest China. *Exposure and health*, 12(3), 487-500.

Received on: 01.02.2022; Revised on: 21.03.2022; Accepted: 23.03.2022

GUIDE FOR AUTHORS

The Journal of Indian Geophysical Union (J-IGU), published bimonthly by the Indian Geophysical Union (IGU), is an interdisciplinary journal from India that publishes high-quality research in earth sciences with special emphasis on the topics pertaining to the Indian subcontinent and the surrounding Indian Ocean region. The journal covers several scientific disciplines related to the Earth sciences such as solid Earth geophysics, geology and geochemistry, apart from marine, atmosphere, space and planetary sciences. J-IGU welcomes contributions under the following categories:

- Research papers and short notes reporting new findings.
- Review articles providing comprehensive overview of a significant research field.

In addition, J-IGU also welcomes short communications, after communications and report on scientific activity, book reviews, news and views, etc.

The manuscript should be submitted electronically as a single word format (.doc file) including the main text, figures, tables, and any other supplementary information along with the signed "Declaration Letter". The manuscript should be submitted by email (jigu1963@gmail.com) to the Chief Editor.

After acceptance of the manuscript the corresponding author would be required to submit all source files (text and Tables in word format) and figures in high resolution standard (*.jpg, *.tiff, *.bmp) format. These files may be submitted to J-IGU as a single *.zip file along with the "Copyright Transfer Statement".

IMPORTANT INFORMATION

Ethics in publishing

J-IGU is committed to ensuring ethics in publication and takes a serious view of plagiarism including self-plagiarism in manuscripts submitted to the journal. Authors are advised to ensure ethical values by submitting only their original work and due acknowledgement to the work of others used in the manuscript. Authors must also refrain from submitting the same manuscript to more than one journal concurrently, or publish the same piece of research work in more than one journal, which is unethical and unacceptable. Editor of J-IGU is committed to make every reasonable effort to investigate any allegations of plagiarism brought to his attention, as well as instances that come up during the peer review process and has full authority to retract any plagiarized publication from the journal and take appropriate action against such authors if it is proven that such a misconduct was intentional.

Similarly, Editor and Reviewers are also expected to follow ethical norms of publishing by ensuring that they don't use any unpublished information, communicated to them for editorial or review purpose, in their own research without the explicit written consent of the author. They are also expected to keep manuscript/ data/ observations/ any other information related to the peer review confidential to protect the interest of the authors. Reviewers should refrain from reviewing the manuscripts in which they have conflicts of interest resulting from competitive, collaborative, or other relationships or connections with any of the authors, companies, or institutions connected to the manuscript.

Conflict of interest

All authors are requested to disclose any actual or potential conflict of interest including any financial, personal or other relationships with other people or organizations within three years of beginning the submitted work that could inappropriately influence, or be perceived to influence, their work.

Submission declaration

Submission of a manuscript implies that the work has not been published previously and it is not under consideration for publication elsewhere, and that if accepted it will not be published elsewhere in the same or any other form, in English or in any other language, without the written consent of the publisher. It also implies that the authors have taken necessary approval from the competent authority of the institute/organization where the work was carried out.

Copyright

After acceptance of the manuscript the corresponding author would be required to sign and submit the "Copyright Transfer Statement".

MANUSCRIPT PREPARATION

The corresponding author should be identified (include E-mail address, Phone/Mobile number). Full affiliation and postal address must be given for all co-authors.

Abstract:

An abstract of not more than 300 words must be included.

Text:

The manuscript should be structured to include a front page containing the title, Author(s) name, affiliation and address of the institute, where

the work was carried out, a short title, and 5-to-6 Key words. Author(s) present address, if different from the above mentioned address, may be given in the footnote. The corresponding author should be identified with an asterisk and his/her email ID should be provided. This page should be followed by the main text consisting of Abstract, Introduction, Methods/ Techniques/ Area description, Results, Discussion, Conclusions, Acknowledgements, and References. Tables and Figures with captions should be inserted at the end of main text. It should not be inserted in the body of the text.

Figures/ Illustrations:

All figures should be provided in camera-ready form, suitable for reproduction (which may include reduction) without retouching. Figures in high-resolution (at least 300 dpi) standard formats (*.jpg, *.tiff, *.bmp) are acceptable. Figures should be numbered according to their sequence in the text. References should be made in the text to each figure. Each figure should have a suitable caption.

Tables:

Authors should take note of the limitations set by the size and layout of the journal. Table should not exceed the printed area of the page. They should be typed on separate sheets and details about the tables should be given in the text. Heading should be brief. Large tables should be avoided and may be provided as supplementary information, if required.

Equations:

Equations should be numbered sequentially with Arabic numerals and cited in the text. Subscripts and Superscripts should be set off clearly. Equation writing software that presents each equation as an object in MS Word will be accepted. Style and convention adopted for the equations should be uniform throughout the paper.

References:

All references to publications cited in the main text should be presented as a list of references in order following the text and all references in the list must be cited in the text. References should be arranged chronologically, in the text. The list of references should be arranged alphabetically at the end of the paper.

References should be given in the following form:

Kaila, K.L., Reddy P.R., Mall D.M., Venkateswarlu, N., Krishna V.G. and Prasad, A.S.S.R.S., 1992. Crustal structure of the west Bengal Basin from deep seismic sounding investigations. *Geophys. J. Int.*, 111,45-66.

REVIEW PROCESS:

All manuscripts submitted to the journal are peer-reviewed. It is advisable to send the contact details of 4 potential reviewers along with the manuscript to expedite the review process. Editor has the option to select reviewers from the list or choose different reviewers. The review process usually takes about 3 months. All enquiries regarding the manuscript may be addressed to the Editor.

GALLEY PROOF:

Technical editing of manuscripts is performed by the editorial board. The author is asked to check the galley proof for typographical errors and to answer queries from the editor. Authors are requested to return the corrected proof within two days of its receipt to ensure uninterrupted processing. The editor will not accept new material in proof unless permission from the editorial board has been obtained for the addition of a "note added in proof". Authors are liable for the cost of excessive alterations to galley proof.

PUBLICATION CHARGES:

There are no page charges for publication and printing charges for b/w figures. However, in view of substantial cost involved in printing of color figures, author will be charged for printing of pages containing color figures @ Rs. 2,500/- per page. The charges may be revised at any time based on cost of printing and production. Author will receive an estimate/ invoice of the color figures reproduction cost along with the galley proof. It is the responsibility of the author to remit the color figures reproduction cost within one month of the receipt of the estimate/invoice.

The corresponding author will receive a soft copy (pdf format) of his/her published article. Should the author desire to purchase reprints of his/her publication, he/she must send the duly signed Reprint Order Form (accompanies the galley proof and contains price details) along with the corrected galley proof to the Editor. The reprint charges must be paid within one month of sending the Reprint Order Form.

Any payment related to printing of color figures and/or purchase of reprints should be made in the form of a Demand Draft in the name of Treasurer, Indian Geophysical Union, payable at Hyderabad.

You may download the pdf file from: <http://www.j-igu.in/IGU-Guide-forAuthors.pdf>